TEOREMA di TAYLOR

Se $f: \mathbb{R}^N \to \mathbb{R}$, $f \in \mathcal{C}^k(A)$ con $A \subseteq \mathbb{R}^N$ aperto, dato $\underline{x}_0 \in A$

Per $\underline{x} \in B(\underline{x}_0, r)$ $\quad r > 0$, $r \in \mathbb{R}$, si ha:

$$f(\underline{x}) = \underbrace{\boxed{\sum_{|\alpha| \le k} \frac{\partial^\alpha f(\underline{x}_0)}{\alpha!} \cdot (\underline{x} - \underline{x}_0)^\alpha}}_{\text{POLINOMIO di TAYLOR}} + \sum_{|\alpha| = k} \underbrace{R_\alpha(\underline{x})}_{\text{RESTO}} \cdot (\underline{x} - \underline{x}_0)^\alpha$$

$\alpha! = \alpha_1! \alpha_2! \dots \alpha_m!$

$\alpha = (\alpha_1, \alpha_2, \dots, \alpha_m)$

$$\lim_{\underline{x} \to \underline{x}_0} R_\alpha(\underline{x}) = 0$$

ESEMPIO: $f(x,y) = x^2 + ye^x$, $\underline{k = 2}$ ORDINE II

$\underline{x}_0 = \underline{(0,0)}$ $\quad \alpha = (\alpha_1, \alpha_2) \longrightarrow \underline{(0,0)}, \underline{(0,1)}, \underline{(1,0)}, \underbrace{(1,1), (2,0), (0,2)}$

$(1,1)$

$$f(\underbrace{x,y}_{\underline{x} - \underline{x}_0:\ (x-0,\, y-0)}) \cong f(0,0) + \frac{\partial f}{\partial x}(0,0) \cdot \frac{1}{1!0!} \cdot (x-0)^1 + \frac{\partial f}{\partial y}(0,0) \cdot \frac{1}{0!1!} \cdot (y-0)^1$$

$\frac{\partial f}{\partial x} \to 2x + ye^x$, $\frac{\partial f}{\partial y} \to e^x$

$$+ \frac{\partial^2 f}{\partial x \partial y}(0,0) \cdot \frac{1}{1!1!} \cdot (x-0)^1 (y-0)^1 + \frac{\partial^2 f}{\partial y \partial x}(0,0) \cdot \frac{1}{1!1!} \cdot xy$$

$\frac{\partial^2 f}{\partial x \partial y} \to e^x$, $\frac{\partial^2 f}{\partial y \partial x} \to e^x$

$$+ \frac{\partial^2 f}{\partial x^2}(0,0) \cdot \frac{1}{2!0!} (x-0)^{\boxed{2}} + \cancel{\frac{\partial^2 f}{\partial y^2}(0,0)} \cdot \frac{1}{0!2!} \cdot (y-0)^2$$

$\frac{\partial^2 f}{\partial x^2} \to 2 + ye^x$, $\frac{\partial^2 f}{\partial y^2} \to 0$

$$= 0 + \cancel{(2 \cdot 0 + 0 \cdot e^0) \cdot x} + e^0 y + 2e^0 \cdot x \cdot y + (2 + 0 \cdot e^0) \cdot \frac{1}{2} \cdot x^2$$

$$= y + 2xy + x^2 \qquad k = 2$$

$$\boxed{f(x,y) \cong y + 2xy + x^2}$$

ATTORNO a $(0,0)$